AMOD–C power and data conversion

Jason Gallicchio and Claude, rev A, 2026–10–25 github.com/gallicchio/AMOD–and–RFMOD

FOOT

FOOT

c11 555 VCO
OUT 6.9 kHz
CV +–1 V
moves it

c5 SC integrator
phases IO4, IO8
R = 1 / f C
C = 1 nF
IN +–1 V

github.com/gallicchio/
AMOD–and–RFMOD
handbook and generator

c4 SC filter
CLK on IO3
fc = fCLK / 100
to 15 kHz
IN +–1 V

0 to –60 dB
1k 100k

c1 Power
USB–C 5 V only
+5 V 1.5 A
–5 V 40 mA
+3.3 V 300 mA

c6 Delta–sigma
IN +–1.5 V max
CLK IO1, BIT IO7
needs gateware
INT1, INT2 / 4

c10 Cascode
jumper on: CE
off: cascode
IN +–50 mV

c3 Charge pump
IO6 clocks it

OUT about –5 V

R = 1 / f C

c2 Regulators
buck 3.6 V 1.5 MHz
LDO 3.3 V
LOAD 100 mA/V,
1 V max
TRIM +–2 V

c8 Comparator
BIT on IO2
high: IN > DAC
SAR in gateware

c7 Track/hold
IO5 high: track
IO5 low: hold
IN +–1 V
droop 30 mV/s

c12 Rectifier
OUT = |IN|
DC to 100 kHz
IN +–1 V

c9 Differential pair
IN+, IN– +–0.2 V
linear +–25 mV
OUT +–1 V

+1 to –1 V
–0.2V +0.2V

FOOT

FOOT

AMOD–C: power and data conversion, switched
capacitors and transistors. USB–C 5 V on c1.

AMOD-C rev A, bottom side seen from below (mirrored), scale 1.90:1